

Responsible & Ethical AI

100 Questions: Ethical Frameworks, Principles, Bias, Governance, Regulation

GARP RAI Exam Preparation

QUESTION: What is Responsible AI?

OPTIONS:

- ❶ A marketing term with no practical meaning
- ❷ The ethical and responsible development, deployment, and use of AI systems to benefit society while mitigating risks
- ❸ A specific type of machine learning algorithm
- ❹ A regulatory framework only applicable in Europe

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Responsible AI aims to ensure AI benefits society holistically while mitigating potential risks. It involves aligning AI systems with ethical standards, industry standards, regulation, legislation, and global principles.
- **Why A is wrong:** Responsible AI has substantive practical meaning and applications
- **Why C is wrong:** It's a governance and ethics framework, not a specific algorithm
- **Why D is wrong:** Responsible AI is a global concept, not limited to Europe

EXAM TRAP: Responsible AI = [ethical development and deployment of AI systems!](#)

Q2: Ethical Frameworks Purpose - Tutorial

QUESTION: What is the purpose of ethical frameworks in AI?

OPTIONS:

- ❶ To make AI development slower
- ❷ To help decide how to design and deploy AI to minimize risks, harms, and wrongs
- ❸ To replace all technical considerations
- ❹ To eliminate all risks completely

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ethical frameworks help define principles to abide by, identify unacceptable biases, assess transparency needs, identify stakeholders, promote accountability, and make justifiable decisions.
- **Why A is wrong:** Ethics aims to guide, not slow, development
- **Why C is wrong:** Technical considerations remain essential
- **Why D is wrong:** Complete risk elimination is impossible

EXAM TRAP: Ethical frameworks [guide decision-making](#) to minimize risks and harms!

Q3: Pluralistic Societies and Ethics - Tutorial

QUESTION: Why are different ethical frameworks important in pluralistic societies?

OPTIONS:

- ① Because everyone agrees on ethics
- ② Because they help us make justifiable decisions when people hold different values
- ③ Because only one framework is ever needed
- ④ Because ethics is not important in diverse societies

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Different ethical frameworks help us make justifiable decisions in pluralistic societies where people hold diverse values and beliefs.
- **Why A is wrong:** People have different ethical views
- **Why C is wrong:** Multiple frameworks provide complementary insights
- **Why D is wrong:** Ethics is particularly important in diverse societies

EXAM TRAP: Different frameworks help make **justifiable decisions** in diverse societies!

QUESTION: What is practical ethics?

OPTIONS:

- ① Theoretical philosophy disconnected from reality
- ② The application of ethical theory and development of good practices to solve real-world moral dilemmas
- ③ A legal framework for AI regulation
- ④ A type of machine learning algorithm

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Practical ethics applies ethical theory to solve real-world moral dilemmas, providing concrete guidance for moral decision making.
- **Why A is wrong:** It is applied, not purely theoretical
- **Why C is wrong:** It's broader than law, informing legal frameworks
- **Why D is wrong:** It's about ethics, not algorithms

EXAM TRAP: Practical ethics = applying theory to real-world dilemmas!

QUESTION: Which is NOT a characteristic of practical ethics?

OPTIONS:

- ① Applied focus on actual ethical dilemmas
- ② Multidisciplinary approach drawing on various fields
- ③ Context-sensitive emphasis on situational nuances
- ④ Ignoring real-world consequences

ANSWER: D

DETAILED EXPLANATION:

- **Why D is correct:** Practical ethics explicitly focuses on real-world consequences and providing concrete guidance, not ignoring them.
- **Why A is wrong:** Applied focus IS a characteristic
- **Why B is wrong:** Multidisciplinary approach IS a characteristic
- **Why C is wrong:** Context sensitivity IS a characteristic

EXAM TRAP: Practical ethics is **applied, multidisciplinary, and context-sensitive** - never ignores consequences!

Q6: Building Trust and Reputation - Tutorial

QUESTION: How does a practical ethics framework help build trust and reputation?

OPTIONS:

- ❶ By making companies look good without real change
- ❷ Customers, employees, and the public are more likely to trust companies known for adhering to ethical standards
- ❸ By hiding unethical practices
- ❹ By avoiding all public scrutiny

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Companies known for ethical standards build trust with customers, employees, partners, and the public, improving brand image and reputation.
- **Why A is wrong:** Ethics requires genuine commitment, not just appearance
- **Why C is wrong:** Hiding practices undermines trust
- **Why D is wrong:** Public scrutiny is part of accountability

EXAM TRAP: Ethics frameworks build **trust and reputation** through genuine adherence to standards!

QUESTION: How can a practical ethics framework help avoid scandals?

OPTIONS:

- ❶ By ignoring potential risks
- ❷ Unethical behavior like fraud can damage reputation; ethics helps prevent major scandals
- ❸ By making scandals less visible
- ❹ By blaming employees for failures

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Unethical behavior (fraud, bias, privacy violations) damages reputation and bottom line. Practical ethics helps prevent major scandals by embedding ethical practices.
- **Why A is wrong:** Ignoring risks increases scandal likelihood
- **Why C is wrong:** Scandals become visible regardless
- **Why D is wrong:** Ethics frameworks promote accountability, not scapegoating

EXAM TRAP: Ethics helps **prevent scandals** by embedding ethical practices!

Q8: Attracting and Retaining Talent - Tutorial

QUESTION: How does a practical ethics framework help with talent?

OPTIONS:

- ❶ By paying higher salaries
- ❷ Workers increasingly care about responsible and ethical practices, so ethics frameworks help recruit and retain talent
- ❸ By reducing workforce requirements
- ❹ By outsourcing ethical decisions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Employees entering the job market are increasingly concerned about their companies' ethical practices. Having a practical ethics framework helps attract and retain talent.
- **Why A is wrong:** Ethics is about values, not just compensation
- **Why C is wrong:** Ethics frameworks don't reduce workforce needs
- **Why D is wrong:** Ethics decisions cannot be fully outsourced

EXAM TRAP: Ethics frameworks help **attract and retain talent** concerned about responsible practices!

QUESTION: How does a practical ethics framework strengthen workplace culture?

OPTIONS:

- ❶ By eliminating all disagreements
- ❷ Ethics programs can nurture teamwork, accountability, and integrity, boosting morale and productivity
- ❸ By enforcing strict rules without flexibility
- ❹ By ignoring employee concerns

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ethics programs nurture teamwork, accountability, and integrity in workplace culture, which in turn boosts morale and productivity.
- **Why A is wrong:** Disagreements are natural and can be healthy
- **Why C is wrong:** Flexibility is important in ethics
- **Why D is wrong:** Employee concerns should be addressed

EXAM TRAP: Ethics frameworks strengthen **teamwork, accountability, and integrity** in workplace culture!

QUESTION: How does a practical ethics framework support risk management?

OPTIONS:

- ① By ignoring all risks
- ② Ethical frameworks can help companies reduce risk of safety failures, data breaches, and regulatory non-compliance
- ③ By transferring all risk to customers
- ④ By avoiding regulation altogether

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Having an ethical framework undergirding policies and procedures can help companies reduce risk of safety failures, data breaches, and regulatory non-compliance.
- **Why A is wrong:** Ignoring risks increases exposure
- **Why C is wrong:** Risk transfer is limited and often unethical
- **Why D is wrong:** Regulation is a reality that must be addressed

EXAM TRAP: Ethics frameworks **reduce risks** of failures, breaches, and non-compliance!

QUESTION: How does an ethical framework guide innovation?

OPTIONS:

- ❶ By stopping all innovation
- ❷ By helping guide innovation in a responsible direction that considers impacts on people and society
- ❸ By making innovation more expensive
- ❹ By ignoring societal impacts

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** An ethical framework helps guide innovation in a responsible direction that considers impacts on people and society, ensuring progress benefits everyone.
- **Why A is wrong:** Ethics enables, not stops, innovation
- **Why C is wrong:** While there may be costs, the benefits outweigh them
- **Why D is wrong:** Societal impacts are central to ethics

EXAM TRAP: Ethics guides innovation in **responsible directions** considering societal impacts!

QUESTION: How does ethics promote long-term thinking?

OPTIONS:

- ① By focusing only on short-term profits
- ② Managing in an environment with ethical considerations helps promote a long-term view
- ③ By avoiding any future planning
- ④ By prioritizing quarterly results

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Managing in an environment with ethical considerations helps promote a long-term view, considering sustainability and future impacts.
- **Why A is wrong:** Ethics balances short and long-term considerations
- **Why C is wrong:** Ethics requires planning for the future
- **Why D is wrong:** Ethics transcends quarterly results

EXAM TRAP: Ethics promotes **long-term thinking** and sustainability!

QUESTION: What is the business case for practical ethics?

OPTIONS:

- ❶ Ethics always reduces profits
- ❷ Proponents argue that practical ethics makes good business sense by providing multiple benefits
- ❸ Ethics is only about avoiding fines
- ❹ Ethics has no business value

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Practical ethics provides multiple business benefits: trust building, scandal avoidance, talent attraction, culture strengthening, risk management, innovation guidance, and long-term thinking.
- **Why A is wrong:** Ethics can enhance long-term profitability
- **Why C is wrong:** Ethics is broader than just avoiding fines
- **Why D is wrong:** Ethics has significant business value

EXAM TRAP: Ethics makes **good business sense** with multiple benefits!

QUESTION: How do ethics and law relate?

OPTIONS:

- ① Ethics and law are the same thing
- ② Ethics is a complement to law, helping ground, inform, and shape law
- ③ Ethics is irrelevant to law
- ④ Law always reflects ethics perfectly

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ethics complements law and is necessary to ground, inform, and shape law. Ethics helps distinguish between just and unjust laws.
- **Why A is wrong:** They are distinct but related
- **Why C is wrong:** Ethics informs law
- **Why D is wrong:** Law often lags behind ethical understanding

EXAM TRAP: Ethics **complements and informs** law!

QUESTION: How does ethics differ from law in scope?

OPTIONS:

- ❶ Ethics is narrower than law
- ❷ Laws establish minimal requirements; ethics goes beyond, identifying moral issues and making recommendations
- ❸ Ethics only applies to individuals, not organizations
- ❹ Law covers all ethical concerns

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Laws establish minimal behavioral requirements. Ethics goes beyond, identifying moral issues, reflecting on the kind of society we want, and making recommendations for a good life.
- **Why A is wrong:** Ethics is broader than law
- **Why C is wrong:** Ethics applies to organizations too
- **Why D is wrong:** Law doesn't cover all ethical concerns

EXAM TRAP: Ethics is **more ambitious than law** - establishes ideals, not just minimums!

QUESTION: What is consequentialism?

OPTIONS:

- ① An ethical theory that judges morality based on adherence to rules
- ② An ethical theory that judges morality based on the consequences of actions
- ③ An ethical theory that focuses on virtuous character traits
- ④ An ethical theory that ignores all outcomes

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Consequentialism judges the morality of an action based on its consequences. The most common form is utilitarianism, which aims to maximize overall utility.
- **Why A is wrong:** This is deontology
- **Why C is wrong:** This is virtue ethics
- **Why D is wrong:** Consequentialism focuses heavily on outcomes

EXAM TRAP: Consequentialism = judging actions by their consequences!

QUESTION: What is utilitarianism?

OPTIONS:

- ① The most common form of consequentialism that aims to maximize overall utility
- ② A deontological theory focused on duties
- ③ A virtue ethics approach to character
- ④ A theory that ignores consequences

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Utilitarianism is the most common form of consequentialism. It aims to maximize overall utility, often defined in terms of pleasure, happiness, or satisfaction of desires.
- **Why B is wrong:** Utilitarianism is consequentialist, not deontological
- **Why C is wrong:** Utilitarianism focuses on consequences, not character
- **Why D is wrong:** Consequences are central to utilitarianism

EXAM TRAP: Utilitarianism = maximize overall utility/happiness!

QUESTION: Who were two famous proponents of utilitarianism?

OPTIONS:

- ① Plato and Aristotle
- ② Jeremy Bentham and John Stuart Mill
- ③ Immanuel Kant and Aristotle
- ④ Confucius and Buddha

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Jeremy Bentham and John Stuart Mill are the two most famous proponents of utilitarianism, developing the theory of maximizing utility.
- **Why A is wrong:** Plato and Aristotle are associated with virtue ethics
- **Why C is wrong:** Kant is associated with deontology
- **Why D is wrong:** These are religious/spiritual figures, not utilitarians

EXAM TRAP: Key utilitarians = **Bentham and Mill!**

QUESTION: What is a key advantage of consequentialism?

OPTIONS:

- ① It provides a single, quantifiable metric for determining moral value
- ② It never leads to difficult decisions
- ③ It always produces the right answer
- ④ It ignores all values

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Consequentialism provides a single, quantifiable metric for determining moral value, making decision-making more straightforward.
- **Why B is wrong:** Consequentialism still involves difficult decisions
- **Why C is wrong:** No ethical theory always produces the right answer
- **Why D is wrong:** It's based on values of utility/happiness

EXAM TRAP: Consequentialism advantage = **single quantifiable metric!**

QUESTION: What is a criticism of consequentialism?

OPTIONS:

- ① It is too focused on rules
- ② It can justify violating individual rights for the greater good
- ③ It ignores consequences
- ④ It is too focused on character

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Critics argue that maximizing utility can lead to actions many consider immoral, like severely violating individual rights for the greater good (e.g., killing one person to save five).
- **Why A is wrong:** Rules are more central to deontology
- **Why C is wrong:** Consequentialism focuses heavily on consequences
- **Why D is wrong:** Character is central to virtue ethics

EXAM TRAP: Consequentialism can justify violating rights for the greater good!

QUESTION: What is rule consequentialism?

OPTIONS:

- ① Judging acts by whether they adhere to utility-maximizing rules
- ② Always breaking rules for better outcomes
- ③ Following all rules regardless of consequences
- ④ Ignoring rules completely

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Rule consequentialists judge acts by whether they adhere to utility-maximizing rules, not only by their case-specific outcomes. This addresses issues with act utilitarianism.
- **Why B is wrong:** This is act utilitarianism, not rule consequentialism
- **Why C is wrong:** This is deontology, not consequentialism
- **Why D is wrong:** This ignores ethical considerations

EXAM TRAP: Rule consequentialism = follow utility-maximizing rules!

QUESTION: What is deontology?

OPTIONS:

- ① An ethical theory that judges morality based on consequences
- ② An ethical framework that judges morality based on adherence to ethical duties and rules
- ③ An ethical theory that focuses on virtuous character traits
- ④ An ethical theory that ignores all moral considerations

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Deontology judges morality based on adherence to ethical duties and rules rather than focusing on consequences. It emphasizes moral obligations regardless of outcomes.
- **Why A is wrong:** This is consequentialism
- **Why C is wrong:** This is virtue ethics
- **Why D is wrong:** Deontology is based on moral considerations

EXAM TRAP: Deontology = judging actions by duties and rules!

QUESTION: Who is the most important proponent of deontology?

OPTIONS:

- ① Jeremy Bentham
- ② Immanuel Kant
- ③ John Stuart Mill
- ④ Aristotle

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Immanuel Kant is the most important proponent of deontology. His categorical imperative states we should act only according to maxims that could be willed as universal law.
- **Why A is wrong:** Bentham is associated with utilitarianism
- **Why C is wrong:** Mill is associated with utilitarianism
- **Why D is wrong:** Aristotle is associated with virtue ethics

EXAM TRAP: Key deontologist = Immanuel Kant!

QUESTION: What is the categorical imperative in Kantian ethics?

OPTIONS:

- ❶ A rule that says consequences determine morality
- ❷ A fundamental principle that one should act only according to maxims that could be willed as universal law
- ❸ A principle that says virtue is most important
- ❹ A rule that says ethics is subjective

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The categorical imperative asserts one should act only according to maxims that could be willed as a universal law without contradiction, such as "It's wrong to lie."
- **Why A is wrong:** This is consequentialism
- **Why C is wrong:** This is virtue ethics
- **Why D is wrong:** Kant believed ethics was objective

EXAM TRAP: Categorical imperative = **act on maxims that could be universal law!**

QUESTION: What does Kant mean by treating people as ends in themselves?

OPTIONS:

- ① Treating people as tools for your purposes
- ② Respecting that people have their own values and goals as autonomous beings
- ③ Using people to achieve your goals
- ④ Ignoring people's autonomy

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Kant argued we shouldn't treat people like things. People are ends in themselves, having their own values and goals as autonomous beings.
- **Why A is wrong:** This is treating people as means, not ends
- **Why C is wrong:** This violates Kant's principle
- **Why D is wrong:** This disregards Kant's emphasis on autonomy

EXAM TRAP: Treating people as ends = [respecting their autonomy and values!](#)

QUESTION: What is a criticism of deontology?

OPTIONS:

- ❶ It focuses too much on consequences
- ❷ It can lead to rigid rule-following even when consequences are disastrous
- ❸ It ignores duties entirely
- ❹ It is too flexible

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Deontology can lead to rigid rule-following even when consequences are disastrous. Kant famously argued lying is always wrong regardless of consequences.
- **Why A is wrong:** Deontology focuses on rules, not consequences
- **Why C is wrong:** Duties are central to deontology
- **Why D is wrong:** Deontology is often criticized as too rigid, not too flexible

EXAM TRAP: Deontology can be **too rigid** even when consequences are disastrous!

QUESTION: What is virtue ethics?

OPTIONS:

- ① An ethical theory that judges morality based on consequences
- ② An ethical framework that judges morality based on adherence to rules
- ③ An ethical theory that emphasizes virtuous character traits and living a good life
- ④ An ethical theory that ignores character

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Virtue ethics emphasizes virtuous character traits and living a good life, rather than focusing on rules or consequences. It asks "What kind of person should I be?"
- **Why A is wrong:** This is consequentialism
- **Why B is wrong:** This is deontology
- **Why D is wrong:** Character is central to virtue ethics

EXAM TRAP: Virtue ethics = focus on character and living a good life!

QUESTION: Who is the ancient Greek philosopher associated with virtue ethics?

OPTIONS:

- 1 Plato
- 2 Aristotle
- 3 Socrates
- 4 Epicurus

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Virtue ethics has roots in ancient Greek philosophers like Aristotle, who taught that happiness comes from living a life guided by virtues.
- **Why A is wrong:** Plato is associated with different philosophical traditions
- **Why C is wrong:** Socrates is associated with different philosophical traditions
- **Why D is wrong:** Epicurus is associated with different philosophical traditions

EXAM TRAP: Key virtue ethicist = [Aristotle](#)!

QUESTION: Which is an example of a virtue in virtue ethics?

OPTIONS:

- ① Maximizing utility
- ② Following all rules
- ③ Wisdom, courage, humanity, justice, temperance, and generosity
- ④ Calculating consequences

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Virtues include wisdom, courage, humanity, justice, temperance, and generosity. These are character traits we should cultivate to live well.
- **Why A is wrong:** This is consequentialism
- **Why B is wrong:** This is deontology
- **Why D is wrong:** This is consequentialist thinking

EXAM TRAP: Virtues include **wisdom, courage, justice, temperance, and generosity!**

QUESTION: What is the key question in virtue ethics?

OPTIONS:

- ① "What action maximizes utility?"
- ② "What duty must I fulfill?"
- ③ "What kind of person should I be?"
- ④ "What are the consequences?"

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Virtue ethics asks "What kind of person should I be?" rather than focusing on universal duties or maximizing utility.
- **Why A is wrong:** This is the consequentialist question
- **Why B is wrong:** This is the deontological question
- **Why D is wrong:** This is consequentialist thinking

EXAM TRAP: Virtue ethics asks "What kind of person should I be?"!

QUESTION: What is a criticism of virtue ethics?

OPTIONS:

- ❶ It provides too much clear guidance
- ❷ It lacks clear guidance for moral decisions compared to duty-based or consequentialist approaches
- ❸ It is too focused on consequences
- ❹ It is too focused on rules

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Critics argue virtue ethics lacks clear guidance for moral decisions compared to duty-based or consequentialist approaches. Different virtues can conflict, and how to weigh them is unclear.
- **Why A is wrong:** It's often criticized for lack of clear guidance
- **Why C is wrong:** Virtue ethics focuses on character, not consequences
- **Why D is wrong:** Virtue ethics focuses on character, not rules

EXAM TRAP: Virtue ethics criticized for **lack of clear guidance!**

QUESTION: How do virtue ethicists respond to criticism about lack of guidance?

OPTIONS:

- ① By ignoring the criticism
- ② By arguing that practical wisdom helps navigate hard cases
- ③ By switching to consequentialism
- ④ By adopting strict rules

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Virtue ethicists counter that practical wisdom helps navigate hard cases. Moral duties can also conflict, and consequences are not always comparable.
- **Why A is wrong:** They engage with the criticism
- **Why C is wrong:** They don't abandon virtue ethics
- **Why D is wrong:** This would be adopting a deontological approach

EXAM TRAP: Virtue ethics relies on **practical wisdom** to navigate hard cases!

QUESTION: How has virtue ethics been extended to organizations?

OPTIONS:

- ❶ It has not been extended
- ❷ Organizational virtues might include justice, accountability, environmental stewardship, and responsible innovation
- ❸ Organizations cannot have virtues
- ❹ Virtue ethics only applies to individuals

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Modern virtue ethics extends beyond individual character. For organizations, virtues might include justice, accountability, environmental stewardship, and responsible innovation.
- **Why A is wrong:** Virtue ethics has been extended to organizations
- **Why C is wrong:** Organizations can embody virtues through their practices
- **Why D is wrong:** Virtue ethics applies to organizations too

EXAM TRAP: Organizational virtues include **justice, accountability, and responsible innovation!**

QUESTION: What is one way virtue ethics applies to AI development?

OPTIONS:

- ❶ By programming all possible rules into AI
- ❷ By having AI learn from experience and habit, like children do
- ❸ By maximizing utility through AI
- ❹ By ignoring character in AI

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Some scholars suggest that to build ethical AI, we must build it in a virtue ethics way, which would imply it learning from experience and habit, like children do.
- **Why A is wrong:** This is a rule-based approach, not virtue ethics
- **Why C is wrong:** This is consequentialism
- **Why D is wrong:** Character is central to virtue ethics

EXAM TRAP: Virtue ethics AI would [learn from experience like children!](#)

QUESTION: How do consequentialism, deontology, and virtue ethics relate?

OPTIONS:

- ① They are mutually exclusive and cannot be combined
- ② They are not mutually exclusive and complement one another in practical ethics
- ③ Only one framework is ever useful
- ④ They all produce identical recommendations

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Consequentialism, deontology, and virtue ethics are not mutually exclusive. In practical ethics, they complement one another. The best moral decision accords with all three theories.
- **Why A is wrong:** They can be combined
- **Why C is wrong:** Each provides valuable insights
- **Why D is wrong:** They have different emphases

EXAM TRAP: Ethical frameworks are **complementary**, not mutually exclusive!

QUESTION: What is the Hippocratic Oath and its significance?

OPTIONS:

- ❶ A modern AI ethics document
- ❷ An ancient Greek oath emphasizing physicians doing no harm to patients
- ❸ A legal document from the 20th century
- ❹ A business ethics code

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The Hippocratic Oath, written between the fifth and third centuries BC, emphasizes the importance of physicians doing no harm to patients. It's one of the earliest medical ethics documents.
- **Why A is wrong:** It's ancient Greek, not modern AI
- **Why C is wrong:** It's from ancient Greece
- **Why D is wrong:** It's medical, not business ethics

EXAM TRAP: Hippocratic Oath = "do no harm" in medicine!

QUESTION: What factors drove the development of medical ethics in the 1970s?

OPTIONS:

- ❶ Only technological advances
- ❷ Concentration of medical care, rising costs, patient rights movements, public outrage at scandals, and rapid technological advances
- ❸ Only government regulation
- ❹ Only public opinion

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Medical ethics evolved in the 1970s due to: concentration of medical care, rising costs, patient rights movements, public outrage at scandals (like Tuskegee), and rapid technological advances.
- **Why A is wrong:** Multiple factors contributed
- **Why C is wrong:** More than just regulation
- **Why D is wrong:** Multiple factors beyond public opinion

EXAM TRAP: Medical ethics evolved due to **multiple societal and technological factors!**

QUESTION: Which is NOT a major medical ethics document?

OPTIONS:

- ① Nuremberg Code (1947)
- ② Declaration of Helsinki (1964)
- ③ Belmont Report (1978)
- ④ EU AI Act (2024)

ANSWER: D

DETAILED EXPLANATION:

- **Why D is correct:** The EU AI Act is an AI regulation, not a medical ethics document. The others are all major medical ethics documents.
- **Why A is wrong:** Nuremberg Code IS a medical ethics document
- **Why B is wrong:** Declaration of Helsinki IS a medical ethics document
- **Why C is wrong:** Belmont Report IS a medical ethics document

EXAM TRAP: EU AI Act is AI regulation, not medical ethics!

QUESTION: Why is medical ethics relevant to AI ethics?

OPTIONS:

- ① They are completely unrelated
- ② Medical ethics has a longer history of addressing practical ethical dilemmas in high-stakes contexts, providing lessons for AI
- ③ Medicine and AI use the same technologies
- ④ Doctors are the only ones who use AI

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Medical ethics has a longer history of addressing practical ethical dilemmas in high-stakes contexts (life and death). AI ethics can learn from these established frameworks and structures.
- **Why A is wrong:** They share important parallels
- **Why C is wrong:** They use different technologies
- **Why D is wrong:** AI is used across many fields

EXAM TRAP: Medical ethics provides **valuable lessons** for AI ethics!

QUESTION: What lesson can AI ethics learn from medical ethics?

OPTIONS:

- ❶ That ethics is unnecessary
- ❷ The importance of established ethical frameworks, structures, and committees to address complex dilemmas
- ❸ That technology always solves ethical problems
- ❹ That only doctors should make ethical decisions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Medical ethics shows the importance of established ethical frameworks, structures (ethics committees), and codes of ethics to address complex dilemmas - lessons AI ethics is still learning.
- **Why A is wrong:** Ethics is essential
- **Why C is wrong:** Technology doesn't solve ethical problems
- **Why D is wrong:** Ethical decisions involve many stakeholders

EXAM TRAP: AI ethics needs **established structures and frameworks** like medical ethics!

QUESTION: What are five common principles in AI ethics codes?

OPTIONS:

- ① Speed, efficiency, profit, growth, and market share
- ② Nonmaleficence, beneficence, justice, autonomy, and explainability
- ③ Privacy, security, transparency, accountability, and fairness
- ④ All of the above are common principles

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The five principles common to many AI ethics codes are nonmaleficence (do no harm), beneficence (do good), justice (fairness), autonomy (respect for persons), and explainability (transparency).
- **Why A is wrong:** These are business metrics, not ethics principles
- **Why C is wrong:** These are related but not the five common principles
- **Why D is wrong:** The specific five are nonmaleficence, beneficence, justice, autonomy, and explainability

EXAM TRAP: Five principles = nonmaleficence, beneficence, justice, autonomy, explainability!

QUESTION: What is the principle of nonmaleficence?

OPTIONS:

- ① The obligation to act for the benefit of others
- ② The obligation to avoid harming others or inflicting injuries
- ③ The obligation to treat people fairly
- ④ The obligation to respect people's autonomy

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Nonmaleficence asserts an obligation to avoid harming others or inflicting injuries, including not imposing unjustified risks of harm.
- **Why A is wrong:** This is beneficence
- **Why C is wrong:** This is justice
- **Why D is wrong:** This is autonomy

EXAM TRAP: Nonmaleficence = "do no harm" - avoid harming others!

QUESTION: What is the principle of beneficence?

OPTIONS:

- ① The obligation to avoid harming others
- ② The moral obligation to act for the benefit of others
- ③ The obligation to treat people fairly
- ④ The obligation to respect people's autonomy

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Beneficence refers to the moral obligation to act for the benefit of others. It requires taking positive steps to help others, not just refraining from harm.
- **Why A is wrong:** This is nonmaleficence
- **Why C is wrong:** This is justice
- **Why D is wrong:** This is autonomy

EXAM TRAP: Beneficence = "do good" - act for the benefit of others!

QUESTION: How do nonmaleficence and beneficence differ?

OPTIONS:

- ① They are the same principle
- ② Nonmaleficence means avoiding harm; beneficence means actively doing good
- ③ Nonmaleficence is more important than beneficence
- ④ Beneficence is a subset of nonmaleficence

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Nonmaleficence is the duty to avoid harming others (do no harm). Beneficence is the duty to actively do good and help others (do good). Beneficence goes beyond mere avoidance of harm.
- **Why A is wrong:** They are distinct principles
- **Why C is wrong:** Both are important in different ways
- **Why D is wrong:** Beneficence is broader than just avoiding harm

EXAM TRAP: Nonmaleficence = avoid harm; Beneficence = do good!

QUESTION: What is the principle of justice in ethics?

OPTIONS:

- ① The obligation to avoid harming others
- ② The obligation to act for the benefit of others
- ③ The moral obligation to act in accordance with principles of fairness, equality, impartiality, and proportionality
- ④ The obligation to respect people's autonomy

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Justice refers to the moral obligation to act in accordance with principles of fairness, equality, impartiality, and proportionality. It requires giving each person their proper due.
- **Why A is wrong:** This is nonmaleficence
- **Why B is wrong:** This is beneficence
- **Why D is wrong:** This is autonomy

EXAM TRAP: Justice = fairness, equality, impartiality, and proportionality!

QUESTION: What is distributive justice?

OPTIONS:

- ① Fair processes and impartiality
- ② Equitable allocation of benefits and burdens in society
- ③ Repairing harms through reconciling victims and offenders
- ④ Respect and fairness between individuals

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Distributive justice focuses on equitable allocation of benefits and burdens in society. It concerns how resources and opportunities are distributed.
- **Why A is wrong:** This is procedural justice
- **Why C is wrong:** This is restorative justice
- **Why D is wrong:** This is interactional justice

EXAM TRAP: Distributive justice = equitable allocation of benefits and burdens!

QUESTION: What is **procedural justice**?

OPTIONS:

- ❶ Equitable allocation of benefits and burdens
- ❷ Fair processes and impartiality
- ❸ Repairing harms through reconciliation
- ❹ Respect and fairness between individuals

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Procedural justice demands fair processes and impartiality. It's about ensuring that the procedures used to make decisions are fair, not just the outcomes.
- **Why A is wrong:** This is distributive justice
- **Why C is wrong:** This is restorative justice
- **Why D is wrong:** This is interactional justice

EXAM TRAP: Procedural justice = **fair processes and impartiality!**

QUESTION: What is the principle of autonomy in ethics?

OPTIONS:

- ① The capacity to make informed, un-coerced decisions about one's life
- ② The obligation to avoid harming others
- ③ The obligation to act for the benefit of others
- ④ The obligation to treat people fairly

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Autonomy refers to the capacity of people to make their own informed, un-coerced decisions about their lives and actions. It commands respecting others' abilities to determine their own course.
- **Why B is wrong:** This is nonmaleficence
- **Why C is wrong:** This is beneficence
- **Why D is wrong:** This is justice

EXAM TRAP: Autonomy = making informed, un-coerced decisions!

QUESTION: How is autonomy applied in healthcare?

OPTIONS:

- ❶ Patients must always follow doctor's orders
- ❷ Patients have a right to voluntary informed consent and refusal regarding their treatment
- ❸ Patients have no say in their treatment
- ❹ Doctors make all decisions for patients

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** In healthcare, respect for autonomy is crucial. Patients have a right to voluntary informed consent and refusal regarding their treatment. Doctors must inform, but patients decide.
- **Why A is wrong:** Patients have autonomy rights
- **Why C is wrong:** Patients have a say in their treatment
- **Why D is wrong:** Shared decision-making is preferred

EXAM TRAP: Autonomy in healthcare = patient consent and refusal rights!

QUESTION: How does autonomy apply to ethical AI?

OPTIONS:

- ❶ AI should make all decisions for people
- ❷ Ethical AI respects people's autonomy by not using coercive or manipulative tactics
- ❸ Autonomy is irrelevant to AI
- ❹ AI should override human decisions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** An ethical AI system respects people's autonomy by not using coercive or manipulative tactics to get people to act in a particular way. Technology should help people further their own goals.
- **Why A is wrong:** This would violate autonomy
- **Why C is wrong:** Autonomy is relevant to AI
- **Why D is wrong:** This would violate autonomy

EXAM TRAP: Ethical AI respects [autonomy by avoiding coercion and manipulation](#)!

QUESTION: What is the principle of explainability in AI?

OPTIONS:

- ① Making AI systems faster
- ② AI systems should provide transparent and understandable explanations for their actions or decisions
- ③ Making AI systems more accurate
- ④ Making AI systems cheaper

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Explainability (or explicability) refers to the idea that AI systems, especially those with decision-making capabilities, should provide transparent and understandable explanations for their actions or decisions.
- **Why A is wrong:** Speed is not explainability
- **Why C is wrong:** Accuracy is not explainability
- **Why D is wrong:** Cost is not explainability

EXAM TRAP: Explainability = transparent and understandable explanations!

QUESTION: Why is explainability important in AI?

OPTIONS:

- ① It makes AI more profitable
- ② It enables accountability, builds trust, and helps identify biases
- ③ It makes AI faster
- ④ It reduces costs

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Explainability is important for accountability (holding designers accountable), building trust (users understand decisions), and identifying biases and errors.
- **Why A is wrong:** Profit is not the primary reason
- **Why C is wrong:** Speed is not the primary reason
- **Why D is wrong:** Cost reduction is not the primary reason

EXAM TRAP: Explainability enables [accountability](#), [trust](#), and [bias detection](#)!

QUESTION: What is local explainability in AI?

OPTIONS:

- ① Explaining the overall behavior of a model
- ② Explaining a specific decision for a single instance or prediction
- ③ Explaining the architecture of a model
- ④ Explaining the training data

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Local explainability focuses on explaining the decisions of a specific AI model on a single instance or prediction. It provides insights into why a particular decision was made.
- **Why A is wrong:** This is global explainability
- **Why C is wrong:** This is model architecture explanation
- **Why D is wrong:** This is data explanation

EXAM TRAP: Local explainability = explaining a single decision/prediction!

QUESTION: What is global explainability in AI?

OPTIONS:

- ❶ Explaining a specific decision for a single instance
- ❷ Looking at an AI model's overall behavior and decision-making processes
- ❸ Explaining the training data
- ❹ Explaining the model's architecture

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Global explainability looks at an AI model's overall behavior and decision-making processes, providing a comprehensive understanding of how the model operates across various inputs.
- **Why A is wrong:** This is local explainability
- **Why C is wrong:** This is data explanation
- **Why D is wrong:** This is architecture explanation

EXAM TRAP: Global explainability = overall model behavior and processes!

QUESTION: What is a counterfactual explanation in AI?

OPTIONS:

- ① An explanation that ignores hypotheticals
- ② A hypothetical scenario that contrasts with the actual decision, showing what would have led to a different outcome
- ③ An explanation that only describes the actual decision
- ④ An explanation that blames the user

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** A counterfactual explanation presents a hypothetical scenario that contrasts with the actual decision. For example, "You would have been approved if you had \$10,000 more in the bank."
- **Why A is wrong:** Counterfactuals are based on hypotheticals
- **Why C is wrong:** It includes alternatives, not just actual decisions
- **Why D is wrong:** It provides actionable insights, not blame

EXAM TRAP: Counterfactual explanations show **what would have led to a different outcome!**

QUESTION: What is bias in the context of AI?

OPTIONS:

- ❶ Random errors in algorithmic predictions
- ❷ A systemic deviation in the output or impact of an algorithm compared to a desired norm or standard
- ❸ When algorithms are always perfectly accurate
- ❹ When algorithms are faster than humans

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Bias in AI refers to a systemic deviation in the output or impact of an algorithm compared to a desired norm or standard. An algorithm is biased when it deviates from its intended function.
- **Why A is wrong:** Bias is systemic, not random
- **Why C is wrong:** Algorithms can be biased
- **Why D is wrong:** Speed doesn't determine bias

EXAM TRAP: AI bias = systemic deviation from desired norm!

QUESTION: Are all biases in AI problematic?

OPTIONS:

- ① Yes, all biases are problematic
- ② No, not all biases are problematic; justifiable biases can form part of a well-designed AI
- ③ Bias never occurs in AI
- ④ Bias is always intentional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Not all biases are problematic from an ethical standpoint. Justifiable biases can form part of a well-designed AI (e.g., prioritizing certain relevant criteria).
- **Why A is wrong:** Some biases are justifiable
- **Why C is wrong:** Bias does occur in AI
- **Why D is wrong:** Bias is often unintentional

EXAM TRAP: Not all biases are **problematic** - some are justifiable!

QUESTION: How can bias arise in problem specification?

OPTIONS:

- ❶ Problems are always specified perfectly
- ❷ Operationalizing complex goals is nuanced, and selected target variables may fail to capture real-world objectives accurately
- ❸ Problem specification never introduces bias
- ❹ All specifications are equally valid

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** An algorithm may exhibit bias from its inception if the goals it is designed to achieve contain inherent problems. Selected target variables may fail to capture real-world objectives accurately.
- **Why A is wrong:** Problems are often specified imperfectly
- **Why C is wrong:** Problem specification CAN introduce bias
- **Why D is wrong:** Specifications vary in quality

EXAM TRAP: Problem specification can introduce **bias through poor operationalization!**

QUESTION: What happened with the algorithm using healthcare expenditure as a proxy?

OPTIONS:

- ❶ It perfectly identified the sickest patients
- ❷ It favored the richest patients who spend more on healthcare, not the sickest ones
- ❸ It was completely unbiased
- ❹ It was never deployed

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** An algorithm meant to identify sicker patients used healthcare expenditure as a proxy, thereby favoring not the sickest patients, but the richest ones who tend to spend more on healthcare.
- **Why A is wrong:** It failed to identify the sickest patients
- **Why C is wrong:** It was biased
- **Why D is wrong:** The example shows it was deployed

EXAM TRAP: Proxies can **introduce bias** by favoring different groups than intended!

QUESTION: What is historical bias in AI?

OPTIONS:

- ❶ Bias that occurs only in the present
- ❷ Bias in data that reflects past discrimination and patterns
- ❸ Bias that is always intentional
- ❹ Bias that never affects algorithms

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Historical bias occurs when ML algorithms rely on historical data that reflects past discrimination and biases (e.g., women historically excluded from banking).
- **Why A is wrong:** Historical bias is about the PAST
- **Why C is wrong:** Historical bias is often unintentional
- **Why D is wrong:** Historical bias DOES affect algorithms

EXAM TRAP: Historical bias = past discrimination reflected in data!

QUESTION: What is the selective labels problem in AI?

OPTIONS:

- ❶ All data has complete labels
- ❷ Historical data rarely shows counterfactual outcomes (e.g., what would have happened if we had hired or lent to different people)
- ❸ Labels are always accurate
- ❹ Labels are never needed

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Historical data rarely shows counterfactual outcomes. A company knows about people it hired or lent to, but not about those it rejected, creating a selective labels problem.
- **Why A is wrong:** Data often has incomplete labels
- **Why C is wrong:** Labels can be inaccurate
- **Why D is wrong:** Labels are needed for supervised learning

EXAM TRAP: Selective labels = **missing counterfactual outcomes** (what would have happened)!

QUESTION: What is sampling bias in AI?

OPTIONS:

- ① When data samples are perfectly random
- ② When the data sample is not random, so trends may not generalize to other populations
- ③ When all groups are equally represented
- ④ When data is always representative

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Sampling bias arises when the data sample is not random. If the sample is not random, the trends shown by the population under study may not generalize to another population.
- **Why A is wrong:** Sampling bias means NOT random
- **Why C is wrong:** This would be balanced sampling
- **Why D is wrong:** Sampling bias means NOT representative

EXAM TRAP: Sampling bias = non-random sample that doesn't generalize!

QUESTION: How can bias emerge in modeling and design?

OPTIONS:

- ❶ Modeling never introduces bias
- ❷ Choices related to optimization functions, regression models, subgroup consideration, and information presentation can all introduce biases
- ❸ Only data introduces bias
- ❹ Only deployment introduces bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Even with sound problem specification and unbiased data, biases can emerge in modeling, validation, and design phases through choices of optimization functions, models, subgroup consideration, and information presentation.
- **Why A is wrong:** Modeling CAN introduce bias
- **Why C is wrong:** Bias can emerge at multiple stages
- **Why D is wrong:** Bias can emerge at multiple stages

EXAM TRAP: Modeling choices can **introduce bias** even with good data!

QUESTION: How can bias emerge during deployment?

OPTIONS:

- ❶ Deployment never introduces bias
- ❷ People may defer entirely to algorithm suggestions, creating unintended effects
- ❸ Algorithms always work as intended in deployment
- ❹ Deployment eliminates all bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Even when all other aspects are addressed, biases can emerge when people defer entirely to algorithm suggestions, creating unintended effects and allowing blame-shifting.
- **Why A is wrong:** Deployment CAN introduce bias
- **Why C is wrong:** Algorithms may not work as intended
- **Why D is wrong:** Deployment doesn't eliminate bias

EXAM TRAP: Deployment can introduce [bias through human deference to algorithms!](#)

QUESTION: Why do people tend to defer to automated systems?

OPTIONS:

- ① Because they are always correct
- ② Because it's convenient, algorithms appear more objective, and it may shield people from responsibility
- ③ Because they are forced to
- ④ Because algorithms are never wrong

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** People defer to automated systems because it's convenient and time-saving, algorithms appear more "objective," and deferring may shield people from responsibility if things go wrong.
- **Why A is wrong:** Algorithms are not always correct
- **Why C is wrong:** They are not forced to
- **Why D is wrong:** Algorithms can be wrong

EXAM TRAP: People defer to algorithms because of **convenience**, **perceived objectivity**, and **responsibility shielding**!

QUESTION: When does algorithmic bias count as discrimination?

OPTIONS:

- ① Always
- ② Never
- ③ When it results in disfavoring people based on protected characteristics (race, sex, ethnicity, age, etc.) in violation of law
- ④ Only when intentional

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Algorithmic bias is likely to lead to discrimination when it results in disfavoring people based on protected characteristics (race, sex, ethnicity, age, etc.) in violation of legal protections.
- **Why A is wrong:** Not all bias is discriminatory
- **Why B is wrong:** Some bias IS discriminatory
- **Why D is wrong:** Discrimination can be unintentional

EXAM TRAP: Discrimination = **disfavoring based on protected characteristics** in violation of law!

QUESTION: What is group fairness?

OPTIONS:

- ① Treating similar individuals similarly
- ② Statistical criteria where, for example, loan demographics mirror the overall population
- ③ Treating every individual uniquely
- ④ Ignoring group-level patterns

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Group fairness involves statistical criteria, such as requiring that the demographics of approved individuals mirror the overall population (e.g., if 51
- **Why A is wrong:** This is individual fairness
- **Why C is wrong:** This is not fairness
- **Why D is wrong:** Group fairness examines group patterns

EXAM TRAP: Group fairness = statistical equality across groups!

QUESTION: What is individual fairness?

OPTIONS:

- ① Statistical criteria across groups
- ② Treating similar individuals similarly
- ③ Ignoring individual differences
- ④ Treating everyone the same regardless of context

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Individual fairness emphasizes treating similar individuals similarly, even though defining similarity can pose challenges.
- **Why A is wrong:** This is group fairness
- **Why C is wrong:** Individual differences matter in fairness
- **Why D is wrong:** Context matters in fairness

EXAM TRAP: Individual fairness = **treating similar individuals similarly!**

QUESTION: Why can't fairness be automated when base rates are unequal?

OPTIONS:

- ① Because algorithms are not powerful enough
- ② Because it's mathematically impossible to satisfy multiple operationalizations of fairness simultaneously when base rates differ
- ③ Because fairness doesn't matter
- ④ Because data is always perfect

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** When base rates between populations are different, it is mathematically impossible to satisfy demographic parity, predictive rate parity, and equal opportunity simultaneously.
- **Why A is wrong:** It's a mathematical impossibility, not a power issue
- **Why C is wrong:** Fairness does matter
- **Why D is wrong:** Data is never perfect

EXAM TRAP: Fairness cannot be automated when **base rates are unequal** - mathematical impossibility!

QUESTION: What is procedural fairness in AI?

OPTIONS:

- ① Ensuring fair outcomes only
- ② Ensuring fair processes and impartial procedures
- ③ Ignoring processes entirely
- ④ Focusing only on consequences

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Procedural fairness provides reassurances that fair outcomes will be sought through impartial and just processes. It's about having the right structures in place.
- **Why A is wrong:** Both process and outcome matter
- **Why C is wrong:** Processes are important
- **Why D is wrong:** Both process and consequences matter

EXAM TRAP: Procedural fairness = fair processes and impartial procedures!

QUESTION: What is the goal when dealing with algorithmic biases?

OPTIONS:

- ❶ To avoid all biases completely
- ❷ To avoid problematic biases, recognizing that some biases are justifiable
- ❸ To ignore biases entirely
- ❹ To eliminate human judgment

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Since there is no such thing as an "objective" algorithm and fairness cannot be fully automated, the task is not to avoid biases in general, but to avoid problematic biases.
- **Why A is wrong:** Avoiding all biases is impossible and undesirable
- **Why C is wrong:** Biases cannot be ignored
- **Why D is wrong:** Human judgment remains important

EXAM TRAP: Goal = **avoid problematic biases** - some biases are justifiable!

QUESTION: Can technology alone solve bias problems?

OPTIONS:

- ① Yes, technology can solve all bias problems
- ② No, technology rarely solves bias problems on its own; fairness cannot be fully automated
- ③ No technology exists for bias detection
- ④ Bias is not a problem technology can address

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Although technological tools can be helpful, technology rarely solves bias problems on its own. Fairness cannot be fully automated, and AI cannot solve the fairness problems it creates.
- **Why A is wrong:** Technology alone is insufficient
- **Why C is wrong:** Technology does exist for bias detection
- **Why D is wrong:** Technology can help, but not solve alone

EXAM TRAP: Technology alone **cannot solve bias** - fairness cannot be fully automated!

QUESTION: What is likely the best way to identify and correct biases?

OPTIONS:

- ❶ Ignoring them
- ❷ Algorithmic auditing by independent assessors
- ❸ Only using technological tools
- ❹ Removing all features

ANSWER: B

DETAILED EXPLANATION:

- ❶ **Why B is correct:** Algorithmic auditing is likely the best way to identify and correct biases. It includes technological tools and statistical analyses, but also a fresh and diverse look at systems.
- ❷ **Why A is wrong:** Ignoring biases is not a solution
- ❸ **Why C is wrong:** Technology alone is insufficient
- ❹ **Why D is wrong:** Removing features may remove relevant information

EXAM TRAP: Algorithmic auditing is **best way to identify and correct biases!**

QUESTION: How can data quality help avoid biases?

OPTIONS:

- ❶ Data quality is irrelevant to bias
- ❷ Ensuring data is diverse, updated, accurate, representative, and free from past discriminatory tendencies helps avoid biases
- ❸ Only synthetic data can avoid bias
- ❹ Data quality cannot help with bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ensuring data is diverse, updated, accurate, representative, and free from past discriminatory tendencies goes a long way toward avoiding biases.
- **Why A is wrong:** Data quality is important
- **Why C is wrong:** Synthetic data is an option, not the only solution
- **Why D is wrong:** Data quality CAN help

EXAM TRAP: Data quality helps **avoid biases** through diversity and representativeness!

QUESTION: How can ethical committees help with fairness?

OPTIONS:

- ① They are unnecessary
- ② They can discuss possible problematic biases and make decisions about trade-offs considering different ethical frameworks
- ③ They eliminate all biases
- ④ They replace all technical work

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ethical committees provide forums where possible problematic biases can be discussed, and decisions about trade-offs can be made considering consequentialist, deontological, and virtue ethics considerations.
- **Why A is wrong:** They are valuable
- **Why C is wrong:** They can't eliminate all biases
- **Why D is wrong:** They complement technical work

EXAM TRAP: Ethical committees help **discuss and address biases** through diverse perspectives!

QUESTION: What is **privacy** in the context of AI?

OPTIONS:

- ① The ability to control all data
- ② Having privacy to the extent that others don't have access to our personal information
- ③ Having complete freedom from all observation
- ④ The right to be forgotten

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** In the context of AI, someone has privacy with respect to some person or institution and in reference to some personal data point if that person or institution has no access to that personal data point.
- **Why A is wrong:** Privacy is about access, not control
- **Why C is wrong:** Some observation is inevitable
- **Why D is wrong:** Right to be forgotten is part of privacy, not the whole definition

EXAM TRAP: Privacy = **lack of access to personal information!**

QUESTION: What is data security?

OPTIONS:

- ① Doing what is necessary to prevent unauthorized access to data
- ② Ensuring data is always accessible
- ③ Making data public
- ④ Deleting all data

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Data security is doing what is necessary to prevent unauthorized access to data. It includes protecting from ransomware, theft, and corruption.
- **Why B is wrong:** Accessibility must be balanced with security
- **Why C is wrong:** Making data public violates privacy
- **Why D is wrong:** Deleting all data is not data security

EXAM TRAP: Data security = **preventing unauthorized access** to data!

QUESTION: Why is privacy an ethical issue?

OPTIONS:

- ❶ Because it's always illegal
- ❷ Because lack of privacy can lead to wrongs, harms, and risks for individuals, institutions, and society
- ❸ Because privacy is never important
- ❹ Because it only affects corporations

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Privacy is an ethical issue because the lack of it can lead to wrongs, harms, and risks for individuals (discrimination, blackmail), institutions (lawsuits, fines), and society (national security risks).
- **Why A is wrong:** Privacy violations may be legal but still unethical
- **Why C is wrong:** Privacy IS important
- **Why D is wrong:** Privacy affects everyone, not just corporations

EXAM TRAP: Privacy protects against **wrongs, harms, and risks** to individuals, institutions, and society!

QUESTION: Why is personal data described as a "toxic asset"?

OPTIONS:

- ❶ Because it's always worthless
- ❷ It's cheap, easy to mine, and profitable, but exposes companies to risk of hacks, leaks, lawsuits, and expensive management
- ❸ Because it's always harmful
- ❹ Because it's never valuable

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Personal data is described as a toxic but potentially highly valuable asset. It's cheap, easy to mine, and profitable, but it exposes the company to risk of hacks, leaks, lawsuits, and is expensive to manage.
- **Why A is wrong:** It can be valuable
- **Why C is wrong:** It's not always harmful
- **Why D is wrong:** It can be very valuable

EXAM TRAP: Personal data is a **toxic asset** - valuable but risky!

QUESTION: What is data minimization?

OPTIONS:

- ① Collecting as much data as possible
- ② Collecting only the data that is necessary to fulfill a specific purpose
- ③ Collecting no data at all
- ④ Collecting data indefinitely

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Data minimization is the most effective way to protect privacy: collecting only the data that is necessary to fulfill a specific purpose.
- **Why A is wrong:** This increases risk
- **Why C is wrong:** Some data is needed for services
- **Why D is wrong:** Data should be deleted when no longer needed

EXAM TRAP: Data minimization = **collect only necessary data!**

QUESTION: What is the right to be forgotten?

OPTIONS:

- ❶ The right to have all data permanently stored
- ❷ A person's right to have private information removed from Internet searches when data is inadequate, irrelevant, or excessive
- ❸ The right to delete all company data
- ❹ The right to never be remembered

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The right to be forgotten is a person's right to have private information about them removed from Internet searches and other directories when the data is inadequate, irrelevant, or excessive.
- **Why A is wrong:** This is the opposite
- **Why C is wrong:** It applies to personal information
- **Why D is wrong:** It's about removing information, not being completely forgotten

EXAM TRAP: Right to be forgotten = **removal of private information when inadequate or irrelevant!**

QUESTION: What is contextual integrity in privacy?

OPTIONS:

- ❶ Data can be used in any context
- ❷ The use of personal data should adhere to contextual norms of privacy
- ❸ Privacy norms are irrelevant
- ❹ Data should always be shared

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** When people give up their data in a particular context, they have certain expectations about how that data will be used. When transferred to a different context, privacy norms are violated.
- **Why A is wrong:** This violates contextual integrity
- **Why C is wrong:** Privacy norms are important
- **Why D is wrong:** Sharing should respect context

EXAM TRAP: Contextual integrity = [data use should respect context-specific norms!](#)

QUESTION: When should personal data be deleted?

OPTIONS:

- ① Never
- ② As soon as it is no longer necessary
- ③ After 50 years
- ④ Only when requested

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Personal data should be deleted as soon as it is no longer necessary. Routine data deletion protects individuals and keeps data accurate (personal data changes quickly).
- **Why A is wrong:** Keeping data forever creates unnecessary risk
- **Why C is wrong:** 50 years is arbitrary and too long
- **Why D is wrong:** Proactive deletion is better than waiting for requests

EXAM TRAP: Delete data as soon as no longer necessary!

QUESTION: What is a good data security practice?

OPTIONS:

- ❶ Using weak passwords
- ❷ Using strong encryption, thorough anonymization, and differential privacy
- ❸ Sharing data widely
- ❹ Storing data unencrypted

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Good data security practices include using all technical tools available, from strong encryption and passwords to thorough anonymization and differential privacy.
- **Why A is wrong:** Weak passwords compromise security
- **Why C is wrong:** Sharing data widely increases risk
- **Why D is wrong:** Unencrypted data is vulnerable

EXAM TRAP: Data security requires **encryption, anonymization, and differential privacy!**

QUESTION: Which is NOT a key privacy principle?

OPTIONS:

- ① Right to privacy
- ② Data minimization
- ③ Collecting as much data as possible
- ④ Control over data

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Collecting as much data as possible is the opposite of data minimization, which is a key privacy principle. It increases risk and violates privacy best practices.
- **Why A is wrong:** Right to privacy IS a key principle
- **Why B is wrong:** Data minimization IS a key principle
- **Why D is wrong:** Control over data IS a key principle

EXAM TRAP: Collecting as much data as possible **violates privacy principles!**

QUESTION: What makes AI difficult to govern?

OPTIONS:

- ❶ AI is always simple to understand
- ❷ Power asymmetries, lack of transparency, lack of interpretability, lack of ethics structures, lack of regulation, unpredictability, and privacy/copyright concerns
- ❸ AI never needs governance
- ❹ Only technical issues matter

ANSWER: B

DETAILED EXPLANATION:

- ❶ **Why B is correct:** AI governance is challenged by power asymmetries, lack of transparency, lack of interpretability, lack of AI ethics structures, lack of national and international regulation, unpredictability, lack of truth-tracking, and privacy/copyright concerns.
- ❷ **Why A is wrong:** AI is often complex and opaque
- ❸ **Why C is wrong:** AI requires governance
- ❹ **Why D is wrong:** Multiple governance challenges exist beyond technical issues

EXAM TRAP: AI governance faces **multiple interconnected challenges!**

QUESTION: What is a power asymmetry in AI governance?

OPTIONS:

- ❶ When governments are more powerful than tech companies
- ❷ Tech companies are often more powerful than some national governments and regulatory agencies
- ❸ When all stakeholders have equal power
- ❹ When power is never an issue

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Many tech companies at the cutting edge of AI are often more powerful (in wealth, influence, and expertise) than some national governments and regulatory agencies, creating governance challenges.
- **Why A is wrong:** Often tech companies have more power
- **Why C is wrong:** Power is often unequal
- **Why D is wrong:** Power asymmetries ARE an issue

EXAM TRAP: Tech companies may be **more powerful than some governments!**

QUESTION: Why is there a lack of transparency in AI systems?

OPTIONS:

- ❶ All AI systems are fully transparent
- ❷ Most AI systems have been developed by private companies not subject to the same transparency requirements as public institutions
- ❸ Transparency is never needed
- ❹ AI systems are always open-source

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Most AI systems have been developed by private companies that may not be subject to the same transparency requirements as public institutions or universities.
- **Why A is wrong:** Most AI systems lack full transparency
- **Why C is wrong:** Transparency IS needed
- **Why D is wrong:** Most AI systems are proprietary

EXAM TRAP: Private companies are **not subject to same transparency requirements** as public institutions!

QUESTION: Why are neural networks sometimes called "black boxes"?

OPTIONS:

- ① Because they are always transparent
- ② Because often not even computer scientists can be sure of exactly how the model is doing what it's doing
- ③ Because they are simple to understand
- ④ Because they never make mistakes

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Neural networks are called "black boxes" because often not even computer scientists can be sure of exactly how the model is doing what it's doing due to their complexity.
- **Why A is wrong:** They are NOT transparent
- **Why C is wrong:** They are complex, not simple
- **Why D is wrong:** They can make mistakes

EXAM TRAP: Neural networks are **black boxes** - even experts can't fully understand them!

QUESTION: How can proxy variables affect interpretability?

OPTIONS:

- ① They always make models more interpretable
- ② If outside observers don't know what proxies the model is using, it becomes difficult to govern the model
- ③ Proxies never affect interpretability
- ④ Proxies eliminate the black-box problem

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The use of proxies can introduce issues, and if outside observers don't know what proxies the model is using, it is difficult to govern that model.
- **Why A is wrong:** Proxies can reduce interpretability
- **Why C is wrong:** Proxies DO affect interpretability
- **Why D is wrong:** Proxies don't eliminate the black-box problem

EXAM TRAP: Unknown proxies **make governance difficult!**

QUESTION: How does AI ethics compare to medical ethics in terms of structure?

OPTIONS:

- ❶ AI ethics has more structure than medical ethics
- ❷ Medical ethics is supported by bioethicists, ethical codes, and committees; AI ethics lacks similar structures
- ❸ Both have identical structures
- ❹ AI ethics has no need for structures

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Medical ethics is supported by bioethicists, ethical codes, ethics committees, licensing, and required coursework. AI ethics lacks similar structures - no required courses, no licensing, no committees.
- **Why A is wrong:** Medical ethics has MORE structure
- **Why C is wrong:** They are very different
- **Why D is wrong:** AI ethics needs structures

EXAM TRAP: AI ethics lacks the **established structures** of medical ethics!

QUESTION: What is the current state of AI regulation globally?

OPTIONS:

- ❶ Comprehensive global regulation exists
- ❷ There are still few national laws regulating AI (with the exception of China), and no specialized agencies overseeing AI
- ❸ All countries have comprehensive AI laws
- ❹ Regulation is unnecessary

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** At the time of writing, there are still no national laws regulating AI (with the exception of China). There are likewise no specialized agencies overseeing AI.
- **Why A is wrong:** Global regulation is limited
- **Why C is wrong:** Most countries lack AI laws
- **Why D is wrong:** Regulation is needed

EXAM TRAP: Few national AI laws exist (except China) - no specialized agencies!

QUESTION: What is emergent behavior in AI systems?

OPTIONS:

- ❶ Behavior that is always predictable
- ❷ Behavior and properties that were not explicitly coded into the system
- ❸ Behavior that never surprises experts
- ❹ Behavior that is always fully understood

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI systems like neural networks can present emergent behavior and properties that were not explicitly coded into the system, surprising both laypeople and experts.
- **Why A is wrong:** Emergent behavior is UNPREDICTABLE
- **Why C is wrong:** It can surprise experts
- **Why D is wrong:** It may not be fully understood

EXAM TRAP: Emergent behavior = **unexpected capabilities not explicitly coded!**

QUESTION: Why are LLMs not truth-tracking?

OPTIONS:

- ❶ They are designed to always tell the truth
- ❷ They make statistical inferences and probabilistic "guesses" to construct plausible responses, not necessarily truthful ones
- ❸ They never make errors
- ❹ They are always accurate

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** LLMs are based on statistical inferences and probabilistic "guesses" to construct responses. They are designed to give plausible responses, but plausible responses are not necessarily truthful.
- **Why A is wrong:** LLMs are not designed for truth
- **Why C is wrong:** LLMs can make errors
- **Why D is wrong:** LLMs can be inaccurate

EXAM TRAP: LLMs give plausible, not necessarily truthful responses!

QUESTION: What are **privacy** and **copyright** concerns with AI?

OPTIONS:

- ❶ There are no concerns
- ❷ Companies scraped data off the internet without clear claims to that data, violating privacy and copyright
- ❸ All AI training data is public domain
- ❹ Privacy and copyright are not relevant to AI

ANSWER: B

DETAILED EXPLANATION:

- ❶ **Why B is correct:** When companies scrape data off the internet, questions arise about whether they have a claim to that data. Privacy violations and copyright infringement are major concerns.
- ❷ **Why A is wrong:** There are significant concerns
- ❸ **Why C is wrong:** AI training data is not public domain
- ❹ **Why D is wrong:** Privacy and copyright ARE relevant

EXAM TRAP: Scraping data raises **privacy and copyright concerns!**

QUESTION: What is the GDPR?

OPTIONS:

- ❶ A US privacy law
- ❷ The European General Data Protection Regulation, designed to regulate personal data
- ❸ An AI-specific regulation
- ❹ A cybersecurity framework

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The GDPR (General Data Protection Regulation) is a European regulation implemented in May 2018, designed to regulate personal data.
- **Why A is wrong:** It's European, not US
- **Why C is wrong:** It's about privacy, not AI specifically
- **Why D is wrong:** It's a privacy regulation, not just cybersecurity

EXAM TRAP: GDPR = [European General Data Protection Regulation](#)!

QUESTION: What rights does GDPR give to data subjects?

OPTIONS:

- ❶ No rights
- ❷ The right to receive information, access data, request erasure, and object to processing
- ❸ Only the right to access data
- ❹ Only the right to delete data

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Under GDPR, data subjects have rights to receive concise information, access personal data, request erasure, and object to processing for marketing or other purposes.
- **Why A is wrong:** GDPR gives significant rights
- **Why C is wrong:** It includes more than access
- **Why D is wrong:** It includes more than deletion

EXAM TRAP: GDPR rights include **information, access, erasure, and objection!**

QUESTION: Why has the GDPR been effective globally?

OPTIONS:

- ❶ Because it only applies in Europe
- ❷ Because it has extraterritorial jurisdiction, applying to companies outside Europe that offer services to European residents
- ❸ Because all countries adopted it
- ❹ Because it's optional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The GDPR applies not only if the data controller is inside Europe, but also if it's outside the European Economic Area if it offers services to data subjects within the EEA.
- **Why A is wrong:** It has extraterritorial effect
- **Why C is wrong:** Not all countries adopted it
- **Why D is wrong:** It's mandatory for those it applies to

EXAM TRAP: GDPR has [extraterritorial jurisdiction](#) - applies globally to companies serving Europeans!

QUESTION: What is the EU AI Act?

OPTIONS:

- ❶ A privacy regulation
- ❷ A law establishing a common regulatory and legal framework for AI
- ❸ A cybersecurity framework
- ❹ A voluntary guideline

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The EU AI Act is a law establishing a common regulatory and legal framework for AI, approved by the EU Parliament in March 2024 and by the Council in May 2024.
- **Why A is wrong:** It's about AI, not just privacy
- **Why C is wrong:** It's about AI regulation, not just cybersecurity
- **Why D is wrong:** It's legally binding, not voluntary

EXAM TRAP: EU AI Act = **common regulatory framework for AI!**

QUESTION: What is the most comprehensive summary of responsible AI?

OPTIONS:

- ❶ Only technical solutions matter
- ❷ Responsible AI requires ethical frameworks, privacy protection, bias mitigation, explainability, and robust governance and regulation
- ❸ Only regulation matters
- ❹ Ethics is optional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Responsible AI requires a comprehensive approach including ethical frameworks (consequentialism, deontology, virtue ethics), privacy protection, bias mitigation, explainability, and robust governance and regulation.
- **Why A is wrong:** Technical solutions alone are insufficient
- **Why C is wrong:** Ethics and other factors also matter
- **Why D is wrong:** Ethics is essential, not optional

EXAM SUCCESS: Understand **ethical frameworks, principles, governance, and regulation** for responsible AI!